MyTradingPal

Human-Computer Interaction 2025-2026

Phase 2 - Lo-fi prototype and heuristic evaluation

3LEIC03 G05

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Project abridged description

MyTradingPal is a mobile-first investing app that blends AI-driven portfolio management with interactive, character-based learning. Users can invest real money or practice in a simulation mode, guided by AI personas that clearly explain their decisions. The app offers real-time performance tracking, explainable recommendations, customizable risk levels, and strong privacy features like biometrics and public privacy modes. It’s designed to make investing accessible and educational, helping users build confidence and financial literacy through a personalized AI experience.

These three tasks/features were implemented in the app’s Lo-fi prototype using Figma:

- Task #1 - Check Portfolio Performance (Low Complexity);
- Task #2 - Configure & Launch AI Character (Medium-High Complexity)
- Task #3 - Analyze AI Recommendation & Execute Trade (High Complexity).

Prototype’s Wireflow

Check Portfolio Performance

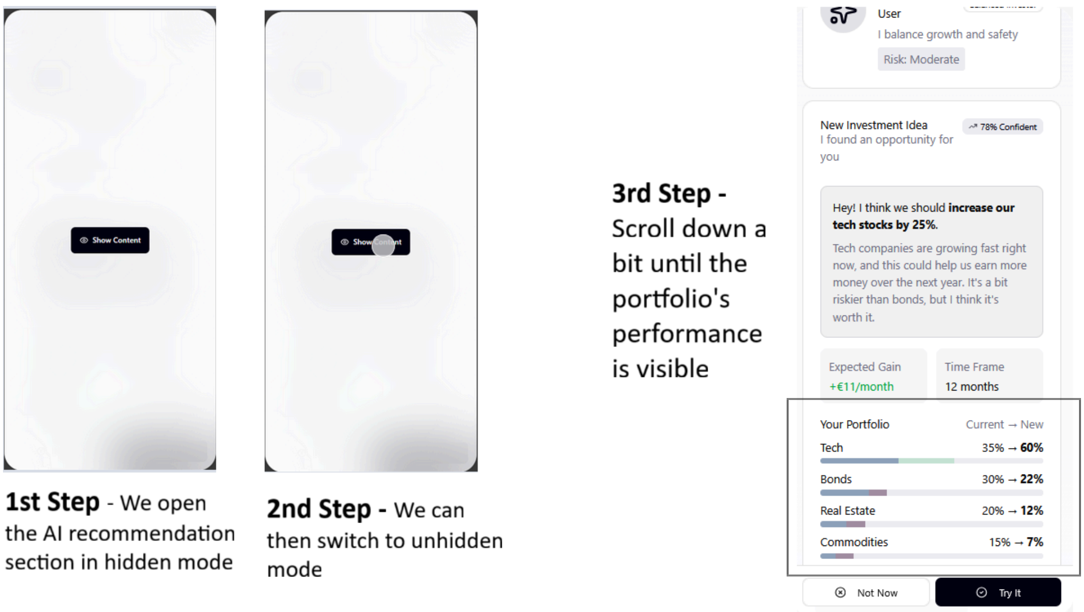
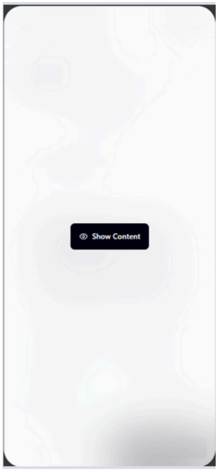


Figure 1 - Task 1

Configure & Launch AI Character



1st Step - We open the AI recommendation section in hidden mode

2nd Step - We can then switch to unhidden mode

Step 3 - Scroll down until the end, click "Choose your persona" followed by the plus (+) sign

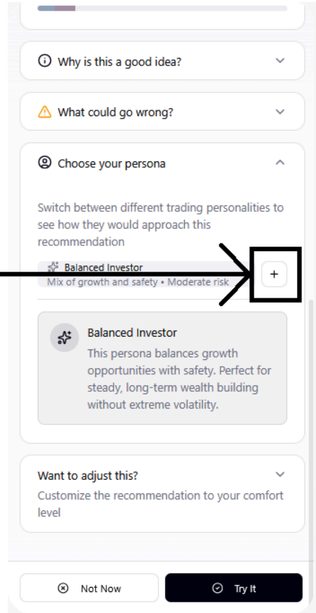
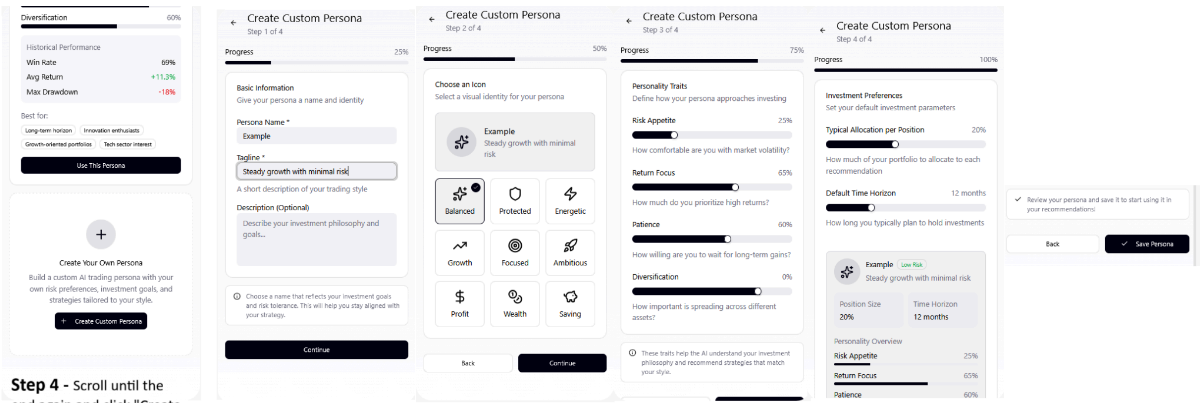


Figure 2 - Task 2 Part 1

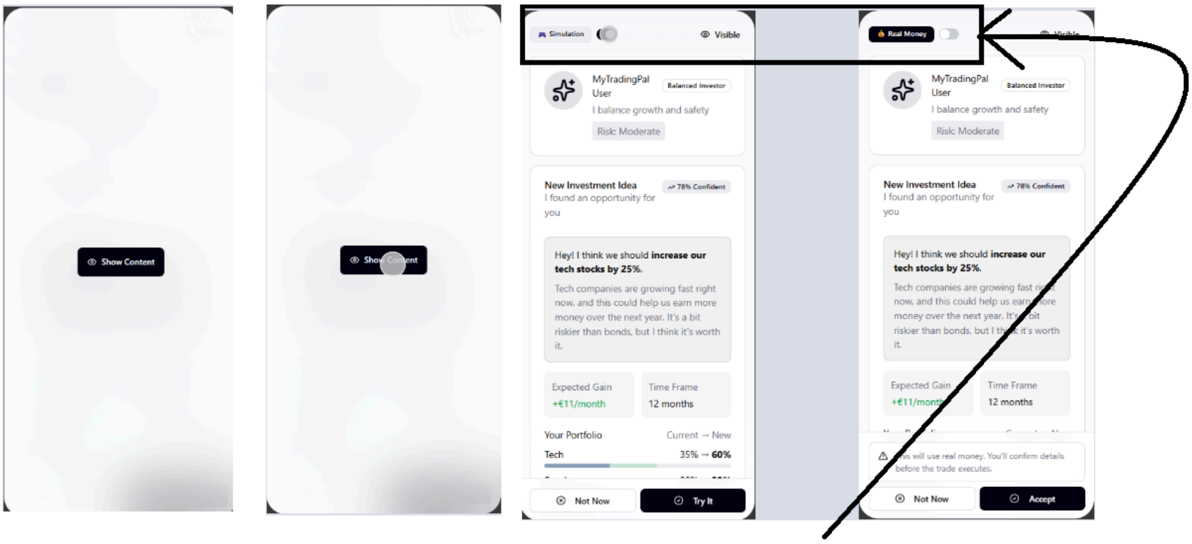


Step 4 - Scroll until the end again and click "Create Custom Persona"

Step 5 - Customize the persona as you see fit. After clicking "Save Persona", it is automatically selected.

Figure 3 - Task 2 Part 2

Analyze AI Recommendation & Execute Trade

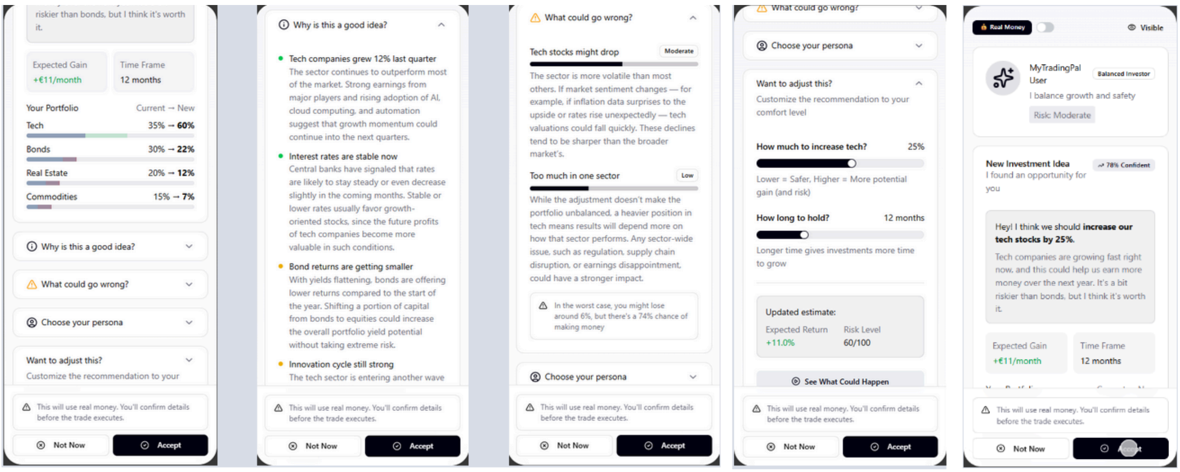


1st Step - We open the AI recommendation section in hidden mode

2nd Step - We can then switch to unhidden mode

3rd Step - After switching to not hidden mode, we can switch between simulation and real world

Figure 4 - Task 3 Part 1



Step 4 - We can now analyze the reasons for the AI recommendation

Step 5 - We can also manually adjust the investment

Step 6 - We can finally either accept or reject the recommendation

Figure 5 - Task 3 Part 2

Heuristic Evaluation Results

Problem #1 - Inefficient Interaction Flow

Identified by group 3, the current design forces the user into an inefficient interaction mode (switching identity) to achieve a simple information goal (comparison). The system makes the user work to access a core value proposition: seeing how personalized the recommendation truly is. This violates Heuristic 7 - Flexibility And Efficiency of Use - and was classified with a severity of 3.

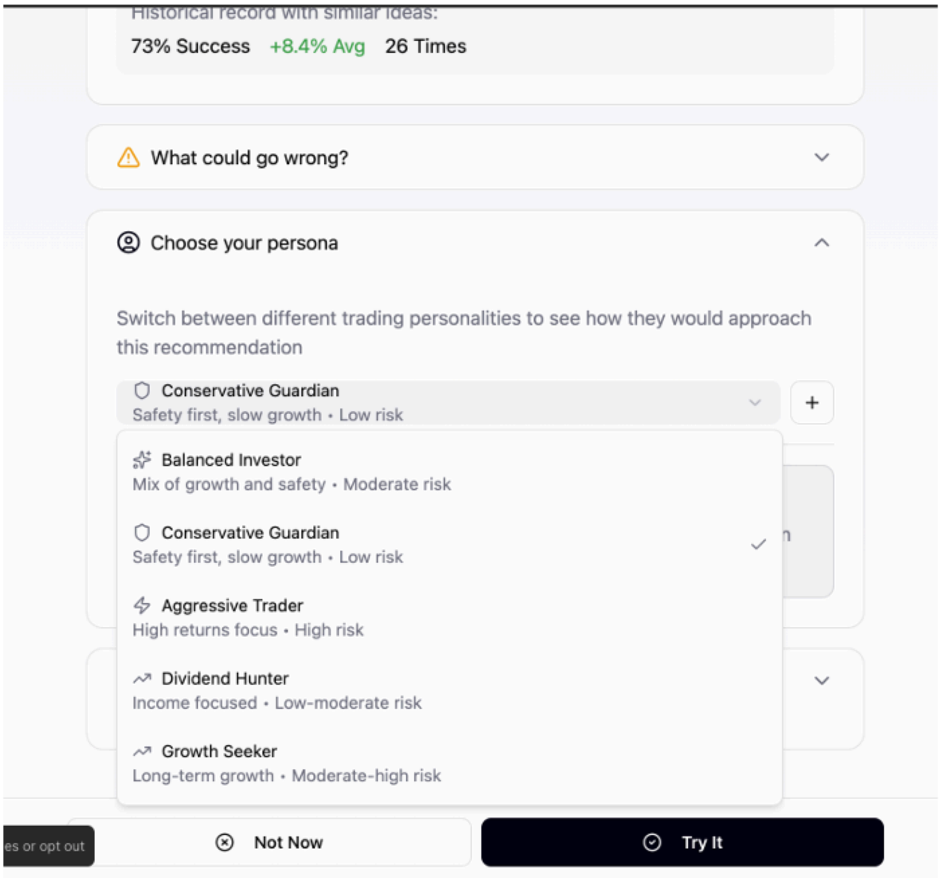


Figure 6 - Inefficient Interaction Flow

As a group, we agree with this observation that the persona-switching flow is inefficient. Requiring users to change identities just to compare recommendations adds unnecessary friction, so we support their assessment.

Problem #2 - Lack of Risk Transparency

Identified by group 3, the specific risk factors (Moderate/Low bars) are presented without a clear link to the overall, quantitative Risk Level (60/100). The system lacks transparency on how the parts contribute to the whole risk score. This violates Heuristic 1 - Visibility of System Status - and was rated with a severity of 2.

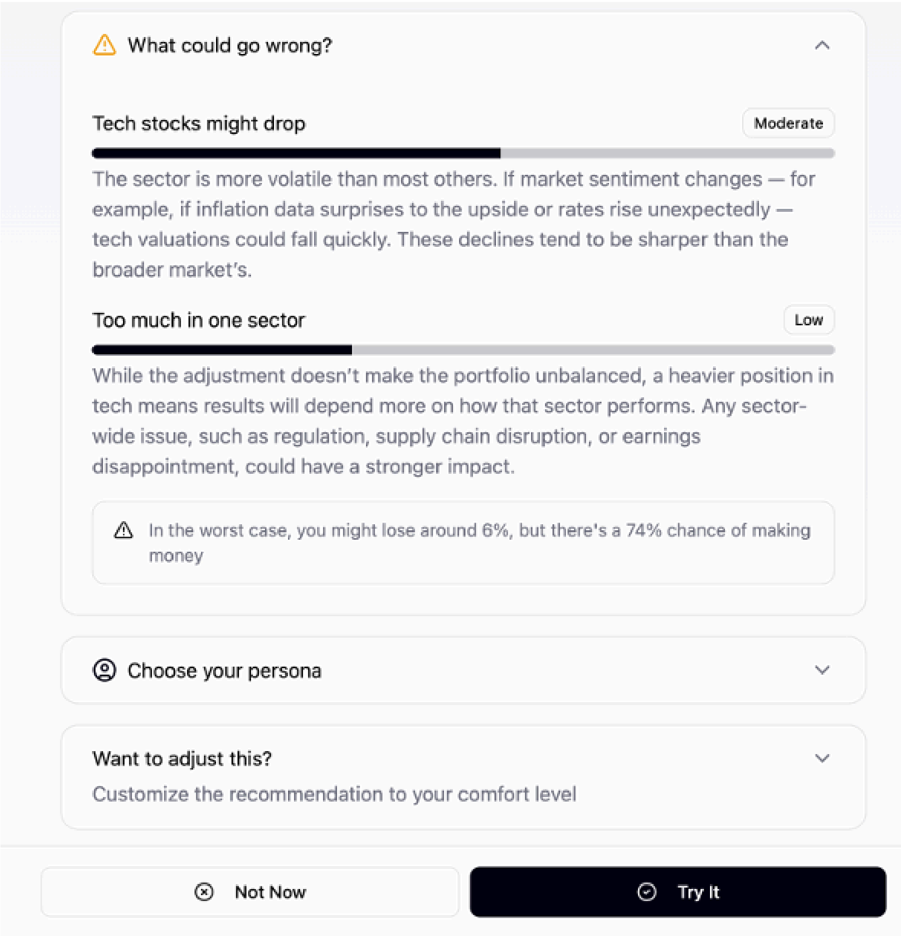


Figure 7 - Lack of Risk Transparency

Our group is in line with group 3's identification of this as a significant usability issue. Their assessment effectively highlights a critical gap in the system's transparency regarding risk calculation and composition.

Problem #3 - Inefficient Persona Comparison

Identified by group 3, to compare the recommendation to another strategy, the user is forced to switch their active profile via a dropdown. This is a cumbersome, multi-step process that hinders efficient exploration and violates Heuristic 7 - Flexibility And Efficiency of Use - being graded with a severity of 3.

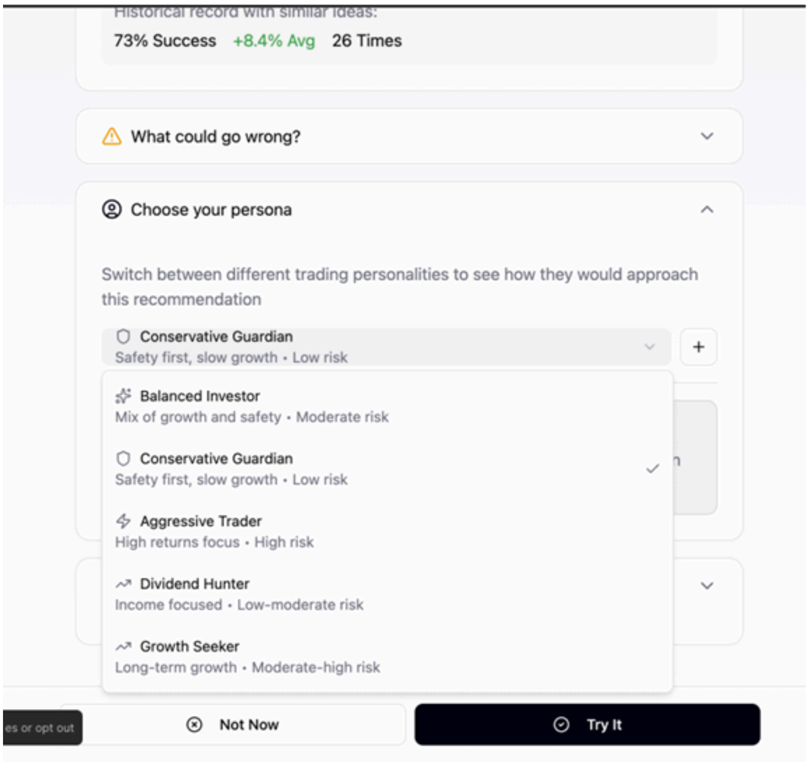


Figure 8 - Inefficient Persona Comparison

Because this problem shares the same root cause as Problem #1, any solution that addresses the inefficient interaction flow would naturally resolve this issue as well. This makes addressing both problems together a logical and efficient approach to improving the system's usability.

Problem #4 - Missing Help and Documentation

Identified by group 4, the application lacks any help button, tutorials, or documentation section. Users have no way to consult when encountering questions about features or investment concepts, such as custom AI creation or risk level interpretation. This violates Heuristic 10 - Help and Documentation and was not classified with a degree of severity.

We agree this is a valid concern. Investment applications inherently involve complex financial concepts, making help resources essential rather than optional. The absence of any guidance mechanism leaves users stranded when questions arise, especially since our app is oriented towards beginners.

Corrections to perform in Phase 3

Solution #1 - Implement side-by-side comparison interface

Introduce a dedicated "Compare" function that enables users to view two AI persona cards side-by-side simultaneously. This solution directly addresses both Problem 1 and 3.

The comparison view would display key information for both personas in parallel, including the specific investment recommendations, persona description, risk level, position allocation, investment timeline, personality traits (Risk Appetite, Return Focus, Patience, Diversification), historical performance metrics (Win Rate, Average Return, Max Drawdown), and suitability tags.

By eliminating the need to switch between personas via the dropdown menu and mentally track differences, users can directly compare how different strategies approach their investment goals. This streamlines the decision-making process and resolves the core inefficiency underlying both identified problems.

Solution #2 - Enhance risk score transparency

Display the proportional contribution of each factor (in ‘What could go wrong?’) to the total score with expandable tooltips explaining the calculation methodology. This solution directly addresses Problem 2 – Lack of Risk Transparency.

The enhanced risk view would show each specific factor alongside its quantitative contribution to the overall Risk Level. Expandable tooltips would provide detailed explanations of the calculations behind each factor, illustrating how individual components combine to produce the total risk score.

By providing a clear mapping between individual risk factors and the overall score, users can understand the system’s scoring logic without guessing or performing mental calculations.

Solution #3 - Develop comprehensive help system

Add a persistent “?” icon accessible from all pages, providing contextual assistance based on the user’s current location within the application. This solution directly addresses Problem 4 – Missing Help and Documentation.

The enhanced help system would provide interactive tutorials for new users, a dedicated FAQ section covering common functionality and terminology questions, hover tooltips for technical or financial terms, and AI-powered assistance that allows users to ask questions in natural language and receive contextual guidance tailored to their current page or task.

By making guidance easily accessible, users can consult the system whenever questions arise about features, such as custom AI creation or investment concepts (like risk level interpretation), without leaving the app. The addition of AI help ensures users receive immediate, personalized explanations, improving understanding and reducing frustration.

Annexes

Sent Heuristic Evaluation Reports

- Group 1: ipc-heuristiceval-T03-G1-G5.pdf
- Group 6: ipc-heuristiceval-T03-G6-G5.pdf

Received Heuristic Evaluation Reports

- Group 3: ipc-heuristiceval-T03-G05-G03.pdf
- Group 4: ipc-heuristiceval-T3G5-G4.pdf